



Largest Submatrix with Non-Repeating Characters

Author : Fares Mezenner

Difficulty: Medium-Hard

Given a matrix m containing ASCII characters (from whitespace " " to "~"), determine the area of the largest submatrix in which no character repeats.

Problem Statement

You are given a $rows \times cols$ matrix where each cell contains a single ASCII character. Your task is to find the largest contiguous submatrix (rectangular region) where all characters are unique and return its area.

Constraints

- Rows and columns smaller or equal 7
- Each cell contains an alphabetic character from the english alphabet, all in lowercase
- The submatrix must be contiguous (i.e., a single rectangular region).

Examples

Example 1

Input: a

Output: 1

Example 2

Input:

e i



B h

Output : 4

Example 3

Input:

e k

g g

e a

Output: 3

Example 4

Input:

r a z a

e v f m

k v m m

v k h e

Output: 6